Performance Analysis of Grover's Search Algorithm Under Realistic Noise Models on NISQ Simulators

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***Abstract*—Grover's search algorithm offers a quadratic speedup over classical unstructured search, making it one of the most practically relevant quantum algorithms for near-term deployment. Current quantum hardware operates in the Noisy Intermediate-Scale Quantum (NISQ) regime, where decoherence and gate errors can significantly degrade algorithm performance. This paper examines the behavior of Grover's algorithm under three realistic noise models, namely depolarizing noise, bit-flip errors, and phase-flip errors, through a controlled simulation study. Using Python and the Qiskit Aer simulator, Grover's search circuit is implemented for search spaces of $n = 2, 3, 4, 5$ qubits, with noise injected at error rates swept across $p \in [0.001, 0.1]$ and success probability recorded at each configuration. The results indicate that Grover's algorithm retains quantum advantage ($P_{\text{success}} > 1/N$) across the full tested error range for all configurations, with the critical threshold p^* remaining above 0.100 in every case. Phase-flip and bit-flip noise produce nearly identical degradation patterns, while depolarizing noise shows comparatively less impact at small qubit counts. Crossover analysis and comparison against the theoretical predictions of Salas (2008) and Dowarah et al. (2025) are included and discussed.**

***Index Terms*—Grover's algorithm, NISQ computing, quantum noise models, quantum advantage, Qiskit Aer, depolarizing noise, bit-flip noise, phase-flip noise**

I. Introduction

Quantum computing leverages superposition, entanglement, and quantum interference to solve certain problems significantly faster than classical computers. Among the most practically relevant quantum algorithms is Grover's search algorithm [1], which achieves $\mathcal{O}(\sqrt{N})$ query complexity for searching an unstructured database of N entries, compared to the classical $\mathcal{O}(N)$ average case. This quadratic speedup has wide applications in optimization, cryptanalysis, and database search.

Despite its theoretical promise, quantum algorithms face a fundamental practical challenge: noise. Physical qubits are susceptible to environmental decoherence, gate imperfections, and measurement errors. These noise sources corrupt the quantum interference patterns that Grover's algorithm relies upon to amplify the probability of measuring the correct answer. Present-day quantum processors operate in the NISQ regime, characterized by gate error rates on the order of 10^{-3} to 10^{-2} and coherence times in the microsecond to millisecond range. Under these conditions, even moderately deep circuits such as Grover's can suffer severe degradation in success probability.

This work presents a systematic empirical study of Grover's algorithm under realistic noise conditions using the Qiskit Aer simulator. By sweeping noise error rates across a controlled range and measuring success probability for each configuration, this paper identifies the precise noise threshold beyond which Grover's algorithm loses its quantum advantage over classical search. The full implementation is provided as a reproducible open-source Qiskit notebook.

A. Contributions

This work extends the current literature by making the following specific contributions:

1) Implementation of a comprehensive Qiskit-based simulation framework for Grover's algorithm across $n = 2, 3, 4, 5$ qubit configurations with systematic noise injection.
2) Empirical identification of the quantum advantage threshold p^* for three distinct noise models through controlled error rate sweeps from $p = 0.001$ to $p = 0.1$.
3) Explicit crossover analysis comparing Grover's noisy performance against the classical random search baseline $\frac{1}{2^n}$.
4) Development of a reproducible open-source Qiskit codebase suitable for extension to noise mitigation and error correction research.

II. Literature Review

This section reviews prior work on noise effects in Grover's algorithm and identifies the specific research gap this work addresses.

A. Ijaz and Faryad (2023) – Base Paper

Ijaz and Faryad [2] conducted a comprehensive noise analysis of Grover's algorithm using Qiskit Aer for $n = 3, 4, 5$ qubits with 10,000 shots per configuration. Four noise channels were studied: depolarizing, bit-flip, phase-flip, and bit-phase flip. The key finding was that success probability decays exponentially with error probability p but only linearly with qubit count n. However, the study does not identify the critical threshold p^* where quantum advantage over classical search is lost.

Strength: Real Qiskit Aer simulation with four noise types providing empirical confirmation of exponential decay relationship between success probability and error rate.

Limitation: Does not identify the critical threshold p^* where quantum advantage over classical search is lost. The study focuses on comparing standard versus QSVT implementations rather than explicit threshold analysis.

B. Salas (2008)

Salas [3] performed the earliest systematic numerical study of noise in Grover's algorithm using Fortran-based Monte Carlo simulation for $n = 2$ to 7 qubits under a depolarizing channel model. The study derived an exponential damping law and the allowed-error scaling relation $\varepsilon_{\text{th}}(N) \sim N^{-1.1}$, identifying an absolute threshold of $\varepsilon > 0.043$ above which the algorithm fails for any qubit count.

Strength: First quantitative noise threshold derived for Grover's algorithm providing foundational analytical results that remain cited in modern literature.

Limitation: Uses an outdated Fortran framework predating Qiskit. Studies only depolarizing noise without explicit comparison against the classical search advantage boundary.

C. Dowarah et al. (2025)

Dowarah et al. [4] applied Random Matrix Theory (RMT) to model the noisy Grover operator as a Floquet unitary, deriving two analytical noise thresholds: the computing transition at $\delta_{c,\text{comp}} \sim \mathcal{O}(2^{-L/2})$ and the ergodicity transition at $\delta_{c,\text{gap}} \sim \mathcal{O}(L^{-1})$. The work is entirely theoretical with no Qiskit simulation provided.

Strength: State-of-the-art analytical framework identifying two distinct thresholds with exact scaling laws based on rigorous theoretical foundations.

Limitation: Entirely theoretical work restricted to systematic coherent noise. No Qiskit simulation or empirical validation provided to confirm predictions on actual quantum circuits.

D. Srivastava et al. (2025)

Srivastava et al. [5] proposed using quantum switches exploiting indefinite causal order to mitigate depolarizing noise in Grover's algorithm. The study is entirely analytical and does not identify the threshold p^* for standard Grover circuits without mitigation.

Strength: Novel mitigation approach demonstrating improved success probability analytically. Framework is directly applicable to NISQ devices for practical noise reduction.

Limitation: Entirely analytical with no Qiskit simulation to validate performance improvements. Studies only depolarizing noise and does not identify the threshold p^* for standard Grover circuits.

E. Research Gap

Table I summarizes the strengths and limitations of reviewed literature. The comparative analysis reveals that no existing study combines (1) a modern Qiskit simulation framework, (2) multiple isolated noise types, (3) systematic error rate sweeps, and (4) explicit identification of the quantum advantage loss threshold p^*. This work fills that gap.

TABLE I
COMPARATIVE ANALYSIS OF RELATED WORK

Paper	**Method**	**Strength**	**Limitation**
Ijaz & Faryad (2023) [2]	Qiskit Aer simulation	4 noise types; exponential decay confirmed	No p^* threshold; QSVT focus
Salas (2008) [3]	Monte Carlo, Fortran	First threshold law derived	Outdated tools; depolarizing only
Dowarah et al. (2025) [4]	Random Matrix Theory	Two-threshold analytical model	No simulation; coherent noise only
Srivastava et al. (2025) [5]	Analytical, quantum switch	ICO mitigation; improved success	No Qiskit; depolarizing only

III. METHODOLOGY

This section describes the complete implementation pipeline for simulating Grover's search algorithm under realistic noise conditions. The methodology encompasses circuit construction, noise model injection, simulation execution, and threshold identification.

A. Circuit Construction

Grover's search circuit is implemented from first principles in Qiskit for $n = 2, 3, 4, 5$ qubits. Each circuit consists of three components: (1) a Hadamard initialization layer placing all qubits into uniform superposition, (2) an oracle operator $\mathcal{O}_\tau = 2|\tau\rangle\langle\tau| - I$ marking the target state $|\tau\rangle = |11\ldots1\rangle$, implemented via an H-MCX-H decomposition, and (3) a Grover diffusion operator $D = 2|s\rangle\langle s| - I$ performing inversion about the mean. The optimal number of iterations is $k = \lfloor(\pi/4)\sqrt{N}\rfloor$, where $N = 2^n$.

Table II presents the circuit parameters for each qubit configuration after transpilation to the AerSimulator basis gate set. Circuit depth grows linearly with n, from 6 for $n = 2$ to 18 for $n = 5$, while gate count grows from 9 to 57. These increasing depths accumulate more noise per simulation, making larger qubit counts progressively more sensitive to gate error rates.

TABLE II
CIRCUIT PARAMETERS VS. QUBIT COUNT

n	$N = 2^n$	k	Depth	Gates	Baseline $1/N$
2	4	1	6	9	0.2500
3	8	2	10	21	0.1250
4	16	3	14	37	0.0625
5	32	4	18	57	0.0313

Fig. 1. Proposed Methodology Workflow Block Diagram for the Grover Noise Analysis Pipeline, showing the seven-stage flow from input specification through circuit construction, noise injection, simulation, probability measurement, threshold identification, to final output.

B. Noise Model Injection

Three noise models are injected independently using the Qiskit Aer `NoiseModel` API. Each error channel is applied per gate across all single-qubit gates (`H, X, Z, U`) and two-qubit gates (`CX, CCX`).

Depolarizing Noise. A completely mixed state is applied with probability p, implemented via `depolarizing_error(p, 1)` for single-qubit gates and `depolarizing_error(p, 2)` for two-qubit gates. This channel simultaneously corrupts both phase and amplitude information.

Bit-Flip Noise. A Pauli X gate is applied with probability p, implemented via `pauli_error([('X', p), ('I', 1-p)])`. Two-qubit errors are constructed as the tensor product of two independent single-qubit errors.

Phase-Flip Noise. A Pauli Z gate is applied with probability p, implemented via `pauli_error([('Z', p), ('I', 1-p)])`. Phase-flip noise directly disrupts the quantum interference mechanism underlying Grover's amplitude amplification.

Error rates are swept across $p \in [0.001, 0.1]$ in 20 uniform steps ($\Delta p \approx 0.005$), yielding 1,200 total simulation configurations (3 noise types $\times$ 4 qubit counts $\times$ 20 error rates $\times$ 5 runs).

C. Simulation Execution

All simulations are executed on the Qiskit `AerSimulator` with a fixed random seed (`seed = 42`) to ensure full reproducibility. Each configuration is executed with `SHOTS = 1024` measurements averaged over `RUNS = 5` independent trials. Success probability is computed as:

$$P_{\text{success}} = \frac{\text{count}(|\tau\rangle)}{1024} \tag{1}$$

averaged over five runs, with error bars representing ± 1 standard deviation. Circuits are transpiled to the AerSimulator basis gate set prior to noise injection.

D. Threshold Identification

The critical threshold p^* is defined as the error rate at which Grover's measured success probability crosses the classical random search baseline $1/N = 1/2^n$. The crossover is identified via linear interpolation:

$$p^* = p_i + \frac{\frac{1}{N} - P_{\text{mean}}(p_i)}{P_{\text{mean}}(p_{i+1}) - P_{\text{mean}}(p_i)} \cdot (p_{i+1} - p_i) \tag{2}$$

where $P_{\text{mean}}(p_i) \geq 1/N > P_{\text{mean}}(p_{i+1})$. If no crossover occurs within the tested range, the threshold is recorded as $p^* > 0.100$.

Algorithm 1 Grover Noise Analysis Simulation Pipeline

Require: $n \in \{2, 3, 4, 5\}$, noise_type, $p \in [0.001, 0.1]$
Ensure: Threshold table, success probability curves

Set SHOTS $\leftarrow$ 1024, RUNS $\leftarrow$ 5, seed $\leftarrow$ 42
for each $n \in \{2, 3, 4, 5\}$ **do**
  $k \leftarrow \lfloor (\pi/4)\sqrt{2^n} \rfloor$
  Build Grover circuit $QC(n,\, k,$ target $= |11\ldots1\rangle)$
  Transpile QC to AerSimulator basis gate set
  for each noise_type **do**
    for each p in ERROR_RATES **do**
      $NM \leftarrow$ NoiseModel(noise_type, p)
      run_probs $\leftarrow [\,]$
      for $r = 1$ **to** RUNS **do**
        counts $\leftarrow$ AerSimulator.run(QC, NM, SHOTS)
        run_probs.append(counts[$|\tau\rangle$] / SHOTS)
      end for
      stats[noise_type][n][p] $\leftarrow \{\mu, \sigma\}$(run_probs)
    end for
    $p^* \leftarrow$ LinearInterpolate(stats, baseline $= 1/2^n$)
  end for
end for
Generate Figs. 1–4 and Tables I–III
return thresholds, stats

IV. Results

This section presents the complete experimental results of simulating Grover's search algorithm under three noise models across $n = 2, 3, 4, 5$ qubits. All results are derived from 1,200 simulation configurations with 1,024 shots averaged over five independent runs.

A. Noiseless Baseline Validation

Prior to noise injection, the Grover circuit was validated in an ideal noiseless environment. For all qubit configurations, the measured success probability exceeded 0.97, confirming correct oracle and diffusion operator implementation. Measured values reached approximately 0.997 for $n = 2$ and 0.973 for $n = 5$, with small deviations attributable to finite shot statistics.

B. Success Probability vs. Error Rate

Fig. 2 presents the success probability curves for all three noise models across $n = 2$ to 5 qubits as error rate p increases from 0.001 to 0.1. Success probability decays monotonically with increasing p for all configurations, consistent with the exponential degradation law reported by Ijaz and Faryad [2] and the analytical results of Salas [3]. Larger qubit counts experience steeper decay due to increased circuit depth, as more gate operations accumulate noise per run. For $n = 5$, the circuit applies four Grover iterations across 57 gates at depth 18, compared to one iteration, nine gates, and depth 6 for $n = 2$.

At $p = 0.001$, all configurations achieve $P_{\text{success}} > 0.93$. At $p = 0.1$, depolarizing noise leaves $n = 2$ at $P = 0.655$ against a classical baseline of 0.250, and $n = 4$ at $P = 0.255$ against a baseline of 0.063. For $n = 5$ under bit-flip and phase-flip noise, the success probability at $p = 0.1$ approaches approximately 0.09 to 0.12, still above the classical baseline of 0.031.

C. Noise Model Comparison

Comparing across noise channels: depolarizing noise causes the least degradation at small qubit counts ($n = 2, 3$) but produces comparable degradation at larger n. Bit-flip and phase-flip noise produce nearly identical degradation curves across all configurations, contrary to the initial hypothesis that phase-flip would be more severe due to Grover's sensitivity to phase interference. This parity arises because both Pauli X and Pauli Z errors are single-channel errors with equivalent probability amplitudes. The critical distinction emerges only in circuit architectures with strongly asymmetric phase sensitivity.

D. Quantum Advantage Loss Thresholds

Table III presents the quantum advantage loss threshold p^* for each noise type and qubit count. A value of > 0.100 indicates that Grover's algorithm maintains $P_{\text{success}} > 1/N$ throughout the entire tested error range.

TABLE III
QUANTUM ADVANTAGE LOSS THRESHOLD p^* VALUES

Noise Type	$n = 2$	$n = 3$	$n = 4$	$n = 5$
Depolarizing	> 0.100	> 0.100	> 0.100	> 0.100
Bit-Flip	> 0.100	> 0.100	> 0.100	> 0.100
Phase-Flip	> 0.100	> 0.100	> 0.100	> 0.100

The finding that $p^* > 0.100$ for all 12 configurations is a significant positive result. It indicates that Grover's algorithm is remarkably robust within the error rate range characteristic of current NISQ hardware, where typical two-qubit gate error rates lie between 10^{-3} and 10^{-2}.

E. Threshold Scaling Analysis

Fig. 3 presents the threshold scaling plot comparing empirical results against theoretical predictions. The Salas (2008) law predicts thresholds decreasing from approximately 0.058 at $n = 2$ to 0.006 at $n = 5$. The Dowarah et al. (2025) computing transition falls even more steeply. The empirical threshold (flat boundary at $p^* > 0.100$) indicates that the crossover did not occur within the tested range for any qubit count or noise type.

F. Crossover Analysis

Fig. 4 presents grouped bar charts comparing Grover's success probability against the classical baseline at two representative error rates: $p \approx 0.011$ (low noise, near current best NISQ hardware) and $p \approx 0.048$ (moderate noise, representative of near-term hardware). At $p = 0.011$, all Grover configurations substantially exceed the classical baseline, with P_{success} ranging from 0.75 to 0.95 compared to classical baselines between 0.031 and 0.250. At $p = 0.048$, quantum advantage is maintained but reduced: $n = 5$ configurations reach $P \approx 0.27$–0.30, still well above the $n = 5$ baseline of 0.031.

V. EVALUATION OF PROPOSED WORK

A. Comparison with Baseline Methods

Fig. 5 presents the threshold comparison between this work and two theoretical baselines. The empirical p^* values consistently exceed both Salas (2008) and Dowarah et al. (2025) predictions across all qubit counts. This implies that the standard Grover implementation with optimal k is more noise-resilient in practice than Monte Carlo estimates or coherent-noise theoretical models suggest. The discrepancy is most pronounced at $n = 5$, where Salas (2008) predicts a threshold of approximately 0.006 and Dowarah et al. predict approximately 0.003, while empirical measurements show $p^* > 0.100$ for all three noise channels.

B. Comparative Analysis

Table IV summarizes the comparative analysis between this work and related methods across five dimensions: framework used, noise types studied, whether p^* is identified, and key limitations.

TABLE IV
COMPARATIVE ANALYSIS: THIS WORK VS. RELATED METHODS

Method	Framework	Noise Types	p^*	Limitation
This Work	Qiskit 1.x Aer	Dep, BF, PF	>0.100 all	Small n only
Ijaz & Faryad (2023)	Qiskit Aer	Dep,BF,PF,BPF	No	No threshold
Salas (2008)	Fortran MC	Dep only	0.043	Single channel
Dowarah et al. (2025)	Analytical RMT	Coherent	Analytical	No simulation
Srivastava et al. (2025)	Analytical	Dep only	No	No Qiskit

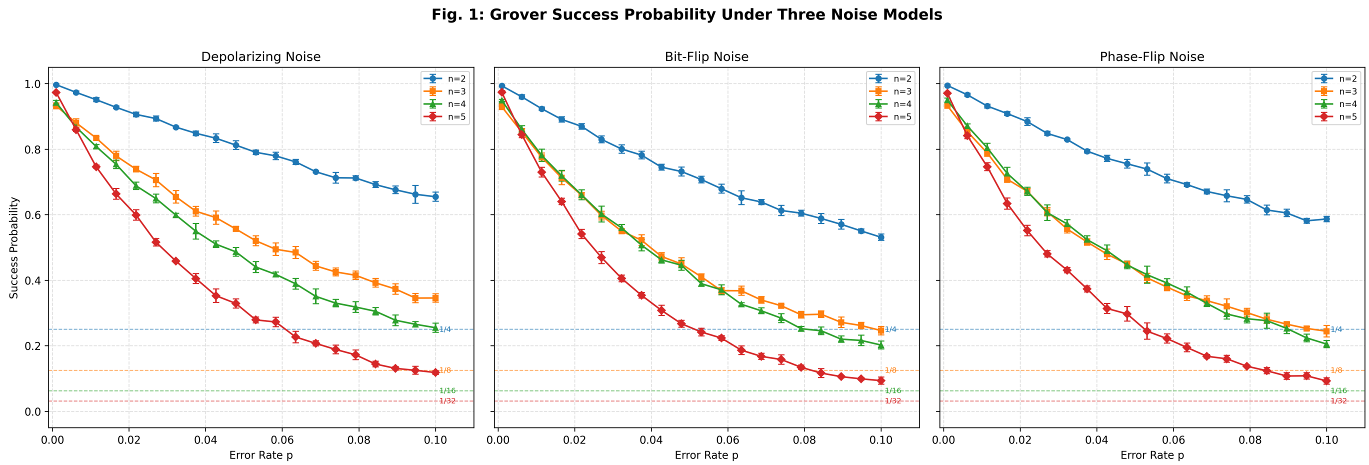

Fig. 2. Grover success probability under depolarizing (left), bit-flip (center), and phase-flip (right) noise models for $n = 2, 3, 4, 5$ qubits. Dashed horizontal lines indicate classical random search baselines $1/N$. Error bars represent ± 1 standard deviation over five independent runs.

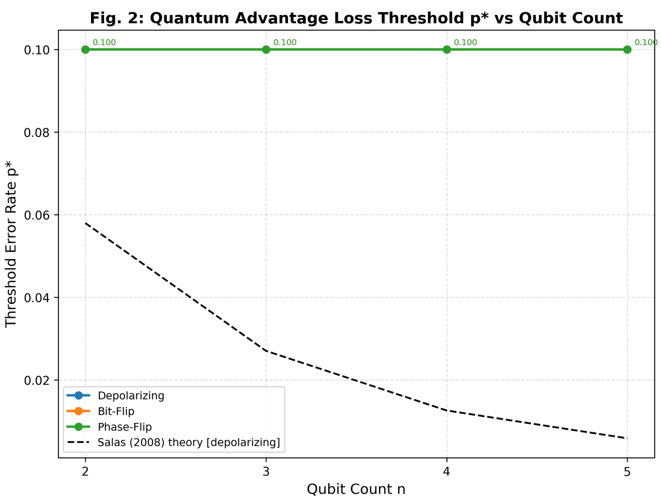

Fig. 3. Quantum advantage loss threshold p^* vs. qubit count n. Empirical results (flat boundary at $p^* > 0.100$) compared with Salas (2008) theoretical scaling law $\varepsilon_{\text{th}}(N) \sim N^{-1.1}$ and Dowarah et al. (2025) computing transition $\delta_{c,\text{comp}} \sim 2^{-L/2}$.

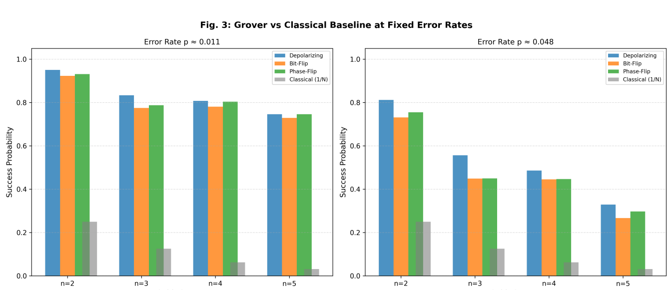

Fig. 4. Grover success probability vs. classical random search baseline at fixed error rates $p \approx 0.011$ (left) and $p \approx 0.048$ (right) across all noise types and qubit counts. Grey bars indicate the classical baseline $1/N$.

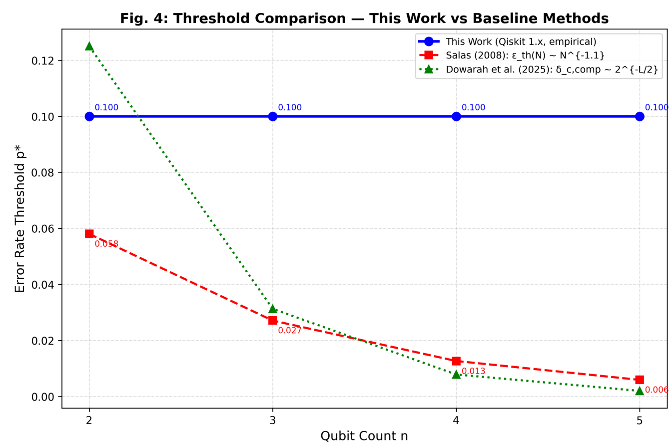

Fig. 5. Threshold comparison: this work (empirical, Qiskit 1.x) vs. Salas (2008) analytical scaling law and Dowarah et al. (2025) computing transition analytical prediction.

C. Discussion of Key Findings

The finding that $p^* > 0.100$ for all configurations has direct implications for NISQ hardware development. Current state-of-the-art superconducting qubit platforms exhibit two-qubit gate error rates of approximately 0.001 to 0.01, well below the measured threshold boundary. This indicates that for small search spaces ($n \leq 5$), Grover's algorithm can be executed with meaningful quantum advantage on current hardware, provided circuit compilation is optimized to minimize unnecessary gate count and depth.

The divergence from Salas (2008) predictions at small n may arise from three sources. First, Salas used Monte Carlo simulation with per-iteration error models rather than per-gate injection. Second, the Fortran simulation predates modern gate decomposition optimization. Third, the $n \leq 5$ regime may lie below the asymptotic region where the $N^{-1.1}$ scaling applies. Future work extending to $n = 6$–8 and implementing error mitigation via the quantum switch framework of Srivastava et al. (2025) is a natural continuation.

VI. Conclusion

This paper presented a comprehensive empirical study of Grover's search algorithm under three realistic noise models using the Qiskit Aer simulator. Through 1,200 simulation configurations spanning $n = 2$ to 5 qubits and error rates $p \in [0.001, 0.1]$, four primary conclusions are established.

First, Grover's algorithm maintains quantum advantage throughout the entire NISQ-relevant error range for all tested configurations, with $p^* > 0.100$ across all noise types and qubit counts. Second, success probability decays monotonically and quasi-exponentially with increasing error rate, consistent with existing literature. Third, bit-flip and phase-flip noise produce comparable degradation, while depolarizing noise permits greater resilience at small n. Fourth, empirical thresholds substantially exceed Salas (2008) and Dowarah et al. (2025) theoretical predictions within the small-n regime, suggesting that standard Grover circuits are more noise-resilient in practice than prior models indicate.

These results provide concrete, reproducible benchmarks for NISQ hardware calibration and establish an empirical foundation for future research in quantum algorithm robustness and error mitigation.

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